

# Anirudh Pilla

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## Summary

Software Development Engineer with 4+ years of experience building scalable, distributed, multi-tenant SaaS platforms. Experienced in microservices, event-driven architecture, and cloud-native systems, delivering production solutions for enterprise clients with focus on performance, reliability, and system design.

## Technical Skills

**Languages:** Python, JavaScript, TypeScript, SQL

**Frameworks:** FastAPI, React, TypeORM, Node.js, NestJS, Express.js, Angular

**Architecture:** Microservices, Event-Driven Architecture, REST APIs, Multi-Tenant Systems, RBAC, SSO

**Databases:** PostgreSQL, MySQL, MongoDB

**Caching & Messaging:** Redis, RabbitMQ, Async Processing Pipelines

**Cloud & DevOps:** AWS, Docker, Jenkins, CI/CD, Grafana

**AI Assisted Tools:** Claude Code, Cursor IDE

**Practices:** System Design, Agile/Scrum, Unit & Integration Testing, Performance Optimization

## Experience

### Akrivia Automation Pvt. Ltd.

Jun 2023 – Aug 2026

Software Development Engineer

Visakhapatnam, India

- Facttwin (Feb 2024 – Aug 2026): Developed and scaled Machine Health Monitoring, a multi-tenant industrial automation SaaS platform supporting 15+ enterprise clients with tenant-level data isolation using NestJS microservices and RabbitMQ-based event-driven architecture and contributed to the SmartFactory project, developing backend services and APIs using Python and FastAPI.
- Improved API performance by 33% by building a centralized NestJS API Gateway with Redis caching, RBAC enforcement, rate limiting, and circuit-breaking mechanisms.
- Reduced client onboarding time by 20% by implementing automated tenant provisioning workflows using RabbitMQ messaging, AES-encrypted communication, and multi-tenant configuration management.
- Designed scalable IoT telemetry processing pipelines using RabbitMQ and Redis, reducing anomaly detection latency while improving data reliability through a dual-database (PostgreSQL + MongoDB) architecture for relational and time-series workloads.
- Akrivia HCM (Jun 2023 – Jan 2024): Developed scalable HR platform modules using NestJS, Angular, MySQL, and Redis, optimizing database queries and API workflows to improve response times by 12%.
- Implemented configurable performance appraisal workflows, including 9-Box evaluation, reducing HR review cycle completion time by 25%.

### Akrivia Automation Pvt. Ltd.

Mar 2022 – May 2023

Software Development Engineer Intern

Visakhapatnam, India

- Akrivia HCM (Mar 2022 – May 2023): Modernized legacy backend and frontend services by migrating to a NestJS + Angular architecture, improving application performance by 15% and reducing maintenance.
- Identified and resolved 30+ production issues in Angular and Hapi.js services during early deployment phases, reducing production incidents and improving overall system reliability.

## Projects

**Cortex** | Python, FastAPI, pgvector, PostgreSQL, OpenAI API, Local LLMs

- Built RAG and tool-calling agent backend to query multi-repository codebases and documentation with real-time streaming. Optimized retrieval accuracy using recursive syntax-aware chunking, pgvector HNSW indexing, and cross-encoder re-ranking to minimize hallucinations. Implemented a custom agentic execution loop for code-symbol searches and integrated a dual-mode fallback supporting local quantized models alongside cloud APIs.

**Boltticket** | Node.js, TypeScript, Redis, BullMQ, PostgreSQL, k6

- Engineered a high-concurrency distributed ticket booking platform sustaining 955 RPS (57K+ requests/min) with no error rate, achieving 3.05 ms median and 7.21 ms P95 latency through Redis Lua-based distributed locking and BullMQ-powered asynchronous workflows.

**LiveBoard** | React, Node.js, Socket.io, HTML5 Canvas, MongoDB

- Built a real-time collaborative whiteboard using React, Socket.io, and HTML5 Canvas, enabling low-latency multi-user drawing, persistent history storage, custom tools, and image export capabilities.

**Face Filters** | MediaPipe, OpenCV, TensorFlow

- Developed a real-time emotion recognition system with 82% accuracy, classifying 7 primary emotions to trigger dynamic facial filters.

## Education

### Raghu Engineering College

Visakhapatnam, India

B.Tech, Computer Science and Engineering (CGPA: 9.16/10)

Mar 2023